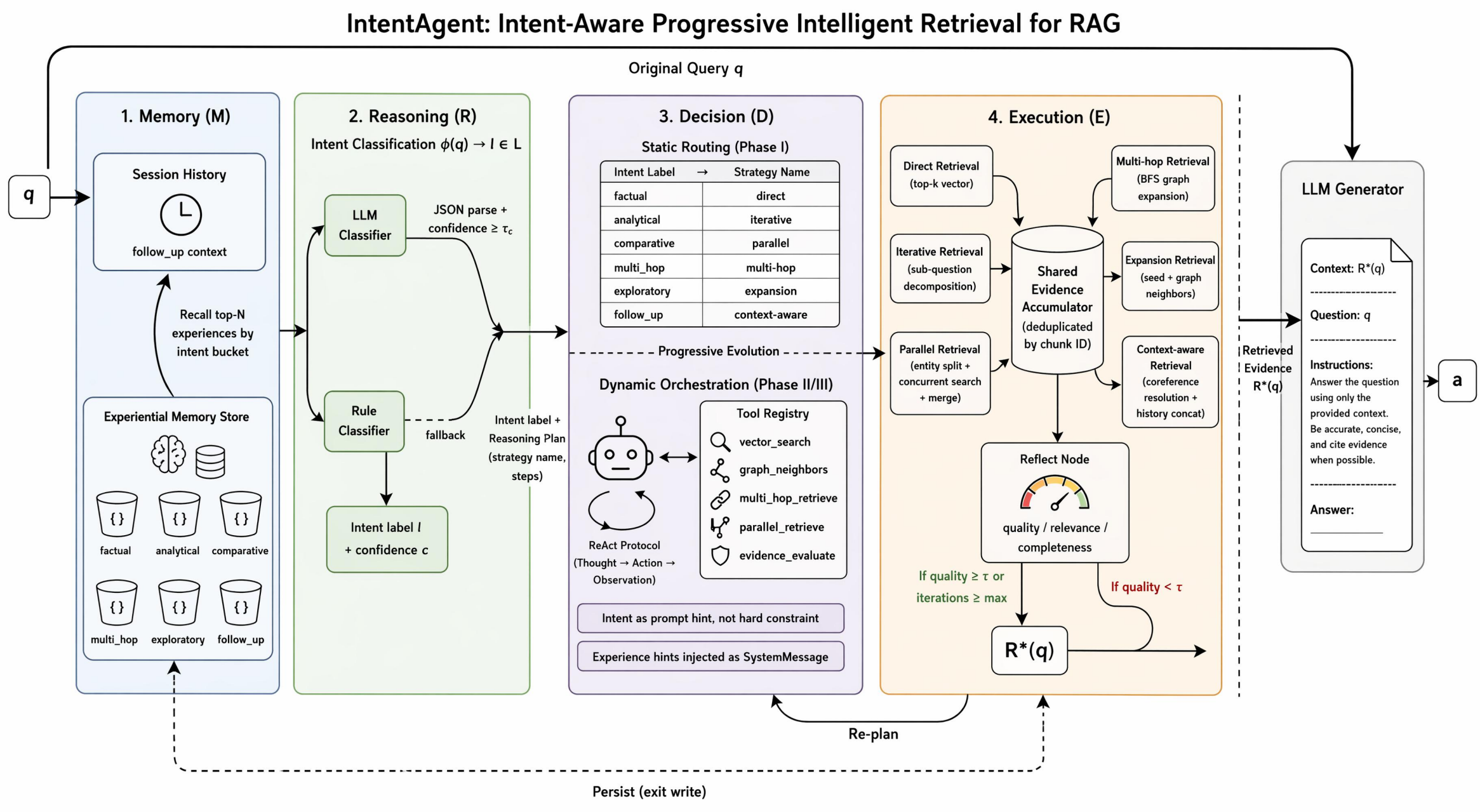

IntentAgent: Intent-Aware Progressive Intelligent Retrieval for RAG
Original Query q
q
1. Memory (M)
Session History
follow_up context
Recall top-N experiences by intent bucket
Experiential Memory Store
factual
analytical
comparative
multi_hop
exploratory
follow_up
2. Reasoning (R)
Intent Classification ϕ(q) → I ∈ L
LLM Classifier
JSON parse + confidence ≥ τc
Rule Classifier
fallback
Intent label I + confidence c
Intent label + Reasoning Plan (strategy name, steps)
3. Decision (D)
Static Routing (Phase I)
Intent Label → Strategy Name
factual → direct
analytical → iterative
comparative → parallel
multi_hop → multi-hop
exploratory → expansion
follow_up → context-aware
Progressive Evolution
Dynamic Orchestration (Phase II/III)
ReAct Protocol (Thought → Action → Observation)
Tool Registry
vector_search
graph_neighbors
multi_hop_retrieve
parallel_retrieve
evidence_evaluate
Intent as prompt hint, not hard constraint
Experience hints injected as SystemMessage
4. Execution (E)
Direct Retrieval (top-k vector)
Multi-hop Retrieval (BFS graph expansion)
Iterative Retrieval (sub-question decomposition)
Shared Evidence Accumulator (deduplicated by chunk ID)
Expansion Retrieval (seed + graph neighbors)
Parallel Retrieval (entity split + concurrent search + merge)
Context-aware Retrieval (coreference resolution + history concat)
Reflect Node
quality / relevance / completeness
If quality ≥ τ or iterations ≥ max
If quality < τ
R*(q)
Re-plan
Retrieved Evidence R*(q)
LLM Generator
Context: R*(q)
Question: q
Instructions:
Answer the question using only the provided context. Be accurate, concise, and cite evidence when possible.
Answer:
a
Persist (exit write)